GRADUATION PROJECT — DEPI ROUND 5

Falcon Team

TechWise

A Smart E-Commerce and Product Intelligence Platform

LITERATURE REVIEW

Intelligent Product Discovery, Explainable Recommendations,
Product Compatibility, Purchase Decision Support, and Verified Second-Life Commerce

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Abstract

Electronic commerce has significantly expanded consumers' access to products, prices, and digital purchasing channels. However, a greater volume of available products does not necessarily translate into better purchasing decisions. Consumers may still face information overload, inconsistent product specifications, limited compatibility information, unexplained recommendations, uncertainty about product condition, and difficulties evaluating whether purchasing a new product, upgrading an existing one, or acquiring a second-hand alternative represents the most appropriate choice.

This literature review examines research relevant to these challenges across six interconnected domains: intelligent product discovery and recommendation, explainable artificial intelligence, product information quality and truthfulness, product compatibility and semantic relationships, multi-criteria purchase decision support, and second-life commerce and product lifecycle traceability. It synthesizes existing findings to identify the technical capabilities, limitations, and design implications relevant to TechWise, a proposed Smart E-Commerce and Product Intelligence Platform.

Existing research provides established foundations for recommender systems, semantic product relationships, information quality assessment, and trust in online second-hand markets. More recent work also investigates large language models as a means of generating recommendation explanations and digital product passports as mechanisms for improving lifecycle information accessibility and traceability. Nevertheless, these research streams address different parts of the product decision and ownership lifecycle, and their integration introduces challenges involving data quality, compatibility validation, explanation fidelity, ownership authorization, privacy, and evidence-based resale eligibility.

Based on this synthesis, TechWise proposes an integrated approach that connects product discovery, structured product intelligence, compatibility evaluation, purchase decision support, and a verified second-life marketplace. Artificial intelligence is positioned as an assistance layer for intent interpretation, requirement extraction, explanation, and recommendation, while deterministic backend services remain responsible for validation, factual product data, compatibility rules, eligibility checks, authorization, and business decisions.

This review establishes an evidence-informed foundation for the proposed platform. It does not claim that the proposed integration has already been implemented, empirically validated, or demonstrated to outperform existing commercial systems.

Keywords: E-Commerce, Recommender Systems, Explainable AI, Product Intelligence, Product Information Quality, Product Compatibility, Multi-Criteria Decision Analysis, Second-Hand Commerce, Product Lifecycle Management, Digital Product Passports.

1. Introduction

1.1 Background

Electronic commerce has evolved from a channel for displaying and selling products into an information-intensive environment in which consumers must evaluate numerous alternatives before making purchasing decisions. Online platforms provide access to product descriptions, technical specifications, prices, images, customer reviews, seller information, and personalized recommendations. These resources can improve product discovery, but their usefulness depends on their relevance, completeness, consistency, and ability to support the consumer's actual decision.

A major challenge in digital shopping is the difference between locating a product and determining whether it is the right product. Search results may contain many technically similar alternatives without establishing which option satisfies a user's requirements. Recommendation systems can reduce the effort required to explore a large catalogue, yet their output may not adequately reflect explicit constraints, technical compatibility, long-term ownership considerations, or the reasons behind a recommendation [1], [2].

Product information introduces another important dimension. Consumers depend on the accuracy and comparability of product attributes when assessing alternatives. Missing specifications, ambiguous descriptions, and inconsistent representations can make comparisons difficult and increase uncertainty. Experimental research on online product information quality has investigated how the quality of product presentation and attributes affects shopping outcomes, emphasizing that information quality is not merely a content-management concern but also a decision-support concern [3].

The decision problem becomes more complex for technology products. A laptop, smartphone, peripheral, or other electronic device may need to satisfy constraints involving performance, budget, connectivity, supported standards, existing equipment, intended workload, and future requirements. A product that performs well in isolation may still be unsuitable for a particular consumer because it does not work with the user's existing equipment or because another alternative provides better overall value.

Furthermore, the purchasing decision does not necessarily end when a transaction is completed. Products may subsequently be maintained, upgraded, repaired, resold, or transferred to another owner. For electronic equipment, information about product history, condition, provenance, and previous ownership can affect the confidence with which consumers evaluate second-hand alternatives. Research on digital product passports and circular electronics identifies product information availability and traceability as relevant considerations for lifecycle decisions, while also recognizing implementation and information-sharing challenges [4], [5].

These challenges suggest that product discovery, evaluation, purchasing, and subsequent ownership should not always be treated as isolated activities. A platform that connects them must

combine intelligent interpretation with reliable product data, explicit business rules, and traceable lifecycle records.

1.2 Problem Context

Many of the challenges relevant to TechWise are addressed individually in existing research:

- Recommender systems support the discovery of potentially relevant products.
- Explainable recommendation methods investigate how systems can communicate reasons for their suggestions.
- Product information quality research examines the usefulness and completeness of information presented to consumers.
- Ontology-based and semantic recommendation methods model relationships between products and user requirements.
- Decision-support methods provide structured approaches to comparing alternatives.
- Second-hand commerce research examines trust, uncertainty, and purchase intentions.
- Product lifecycle and digital product passport research investigates the availability and traceability of product information across lifecycle stages.

These areas provide complementary foundations, but they should not be assumed to form one unified solution automatically. Integrating them requires a consistent representation of products, explicit rules for validating product facts, appropriate handling of uncertainty, and clear separation between generated explanations and authoritative system decisions.

TechWise is proposed in response to this integration challenge. Its purpose is not simply to display products or generate recommendations, but to support a broader product lifecycle:

Discover → Understand → Compare → Decide → Buy → Own → Upgrade → Resell → Transfer

The lifecycle represents the intended scope of the platform. It should not be interpreted as evidence that every stage has already been implemented.

1.3 Aim of the Literature Review

The primary aim of this review is to examine the scientific and technical foundations relevant to the design of TechWise and to identify the research-informed requirements that should guide its architecture, business logic, and future evaluation.

The review focuses on the relationships between recommendation quality, product information reliability, compatibility validation, purchase decision support, and second-life lifecycle traceability.

1.4 Scope

The review considers research related to online retail, recommender systems, explainable artificial intelligence, product information quality, semantic product representations, decision-support

methods, second-hand electronic commerce, and circular product information systems.

The scope is intentionally broader than a review of AI recommendation algorithms alone. TechWise requires a combination of intelligent assistance and conventional software engineering mechanisms. Consequently, the review also considers data governance, authorization, business rules, information provenance, and the distinction between a product model and an individually owned physical asset.

The review does not attempt to establish that every feature of TechWise is novel. Instead, it examines how established approaches can inform the integration of the platform's confirmed core capabilities.

2. Review Objectives and Research Questions

2.1 Review Objectives

This review has five principal objectives.

Objective 1 — Examine intelligent product discovery. Investigate established recommender-system approaches and their suitability for helping consumers identify products that match explicit requirements and inferred preferences.

Objective 2 — Examine trustworthy AI assistance. Analyze the potential role of explainable recommendations and large language models, including the limitations of generated explanations and the need for independent factual validation.

Objective 3 — Examine product information and compatibility. Investigate how structured product attributes, information quality, semantic relationships, and compatibility rules can support reliable product evaluation.

Objective 4 — Examine purchase decisions and second-life commerce. Study how decision-support approaches and trust-related mechanisms can inform purchasing, upgrading, and second-hand acquisition.

Objective 5 — Identify integration gaps. Synthesize findings across these domains to derive an evidence-informed conceptual approach for TechWise and identify the questions that require subsequent implementation and evaluation.

2.2 Research Questions

The review is guided by the following research questions.

ID	Research Question	Primary Research Area
RQ1	Which recommender-system approaches are suitable for personalized product discovery in e-commerce, and what limitations affect their use?	Recommender systems
RQ2	How can AI-generated explanations improve the transparency and usefulness of product recommendations, and what limitations must be	Explainable AI

ID	Research Question	Primary Research Area
	addressed?	
RQ3	How does product information quality affect consumers' ability to evaluate and compare online products?	Product information quality
RQ4	How can semantic relationships and structured technical attributes support product compatibility assessment?	Product intelligence and compatibility
RQ5	How can structured decision-support methods help consumers compare alternatives under multiple requirements and constraints?	Purchase decision support
RQ6	Which factors influence trust in second-hand electronic commerce, and how can product history and traceability support informed decisions?	Second-life commerce
RQ7	What architectural and research challenges arise when recommendation, product truth, compatibility, decision support, and product lifecycle management are integrated into one platform?	Cross-domain synthesis

Table 1. Research questions guiding the TechWise literature review.

RQ7 is a synthesis question. It is addressed by analyzing the relationships and limitations identified in the preceding research areas; it is not a claim that a comprehensive empirical study of all integrated systems has already been conducted.

3. Literature Review Methodology

3.1 Review Design

This document uses a structured, focused literature review to synthesize research relevant to the technical scope of TechWise. It combines foundational research, domain-specific studies, and recent publications addressing developments in recommendation systems, AI-assisted explanations, and product lifecycle information.

The review follows the general principles of transparent literature identification, explicit research questions, consistent comparison criteria, and evidence-based synthesis. Guidelines for systematic literature reviews in software engineering emphasize the importance of a defined protocol and reproducible selection process [6]. PRISMA 2020 provides an additional reporting framework for systematic reviews, including transparent reporting of study identification and selection [7].

However, this document should not be described as a completed systematic literature review or a PRISMA-compliant systematic review. A fully reproducible systematic review would require a documented search across specified databases, recorded search dates and complete queries, duplicate removal, documented screening of all retrieved records, eligibility decisions, and a verified study-selection flow.

The initial research for this draft used exploratory searches of scholarly publishers and accessible research repositories. The sources were selected for their relevance to the project's principal research domains and the availability of identifiable publication information. The selection is intended to establish a credible starting evidence base, not to claim exhaustive coverage of all relevant publications.

3.2 Sources of Literature

The initial source set includes research published through academic publishers and journals, including:

- IEEE Xplore and IEEE Access.
- Frontiers in Big Data.
- Springer Nature journals.
- ScienceDirect and other Elsevier publications.
- Taylor & Francis journals.
- Open-access research repositories such as PubMed Central.

These sources provide access to peer-reviewed articles, review papers, experimental studies, and research concerning product information and lifecycle traceability.

The final document should retain the original publication record, DOI, and official publisher or repository URL wherever available. Search-engine snippets and automatically generated summaries should not replace examination of the underlying paper.

3.3 Search Strategy

The exploratory search used combinations of the following terms:

- "e-commerce" AND "recommender systems" AND "systematic review"
- "explainable recommendation" AND "large language models"
- "product information quality" AND "online shopping"
- "ontology-based" AND "product recommendation"
- "product compatibility" AND "semantic relationships"
- "purchase decision support" AND "multi-criteria"
- "second-hand products" AND "consumer trust"
- "electronic products" AND "digital product passport"
- "product lifecycle traceability" AND "circular economy"
- "repair versus replacement" AND "electronic equipment"

These queries reflect the conceptual scope of TechWise. They should be treated as the initial search strategy rather than a record of a completed exhaustive database search.

3.4 Inclusion Criteria

A source was considered potentially relevant when it met one or more of the following criteria:

1. It examined recommendation methods or product discovery in online commerce.
2. It investigated explainability, transparency, or the role of large language models in recommendation.
3. It provided empirical or conceptual evidence concerning product information quality.
4. It examined semantic or ontology-based product representations and recommendation.

5. It investigated consumer trust in second-hand commerce or product lifecycle information.
6. It addressed product repair, replacement, reuse, or lifecycle decisions relevant to electronic products.
7. It provided a methodological foundation for conducting or reporting a literature review.

3.5 Exclusion Criteria

The intended review excludes sources that lack identifiable publication information, are unrelated to the research questions, make unsupported commercial claims, or provide insufficient evidence for the specific claim being considered.

A source should not be excluded merely because its findings differ from the proposed TechWise approach. Contradictory findings and methodological limitations are important to a balanced academic review.

3.6 Evidence Extraction and Analysis

For each selected publication, the review considers:

- Research objective and problem definition.
- Methodology and type of evidence.
- Main findings relevant to the research questions.
- Evaluation methods, where applicable.
- Limitations acknowledged by the authors.
- Relevance to the proposed platform.
- Implications for system design and future evaluation.

The synthesis distinguishes findings explicitly reported by researchers from interpretations developed for TechWise. Numerical results reported in individual studies are not treated as directly comparable unless their datasets, evaluation conditions, metrics, and experimental settings support such a comparison.

3.7 Methodological Limitations

The current evidence base is a focused selection of relevant publications, not an exhaustive census of research in every domain. The reviewed studies differ in age, scope, research design, datasets, and evaluation methods. Some publications address general recommendation systems, whereas others focus on specific sectors such as clothing retail or electronic products.

The findings therefore support a reasoned design proposal rather than a definitive claim that one integrated approach is universally superior. A subsequent formal review should expand the search, document screening decisions, assess study quality, and verify the final bibliography before submission.

4. Intelligent Product Discovery and Recommendation Systems

4.1 The Role of Recommender Systems in E-Commerce

Recommender systems help users navigate large catalogues by identifying products that may be relevant to their preferences, behavior, or stated needs. They address a central problem of online retail: consumers may face too many alternatives to evaluate efficiently.

Roy and colleagues' systematic study of e-commerce recommender systems classified traditional methods into content-based filtering, collaborative filtering, demographic-based filtering, knowledge-based filtering, and hybrid approaches [1]. The review examined issues such as accuracy, scalability, response time, security, and diversity, illustrating that recommendation quality is multidimensional rather than reducible to a single predictive metric.

Content-based methods evaluate similarities between items and user preferences using product attributes or representations. Collaborative filtering uses patterns of interactions or preferences across users and items. Knowledge-based methods apply explicit knowledge about user requirements and product suitability. Hybrid approaches combine multiple sources of evidence to address limitations associated with individual methods.

These approaches offer different benefits and constraints. Their suitability depends on the availability of product attributes, interaction histories, explicit user constraints, and the quality of the underlying data.

4.2 Content-Based Filtering

Content-based filtering recommends products whose attributes match a user's interests or the attributes of items that the user has previously preferred.

In an e-commerce setting, product representations may include category, brand, price, technical specifications, features, and descriptive text. A recommendation system can compare these attributes with a user profile or a structured representation of the user's current requirements.

The principal advantage of content-based methods is that they can operate without requiring a large population of users who have interacted with the same product. They may therefore be useful when a platform has limited interaction history but possesses structured product data.

However, content-based methods depend heavily on the quality and expressiveness of product representations. A recommendation system may identify products with similar descriptions without establishing whether they satisfy a user's technical requirements. Similarity is not equivalent to suitability.

For TechWise, content-based matching can inform product discovery, but the matching process should distinguish soft preferences from hard constraints. For example, a preference for a particular manufacturer may influence ranking, while an explicitly required connection standard may need to be validated before a product can be considered compatible.

4.3 Collaborative Filtering

Collaborative filtering uses relationships between users, products, or interaction patterns to identify items that may be relevant to a user.

Its principal strength is the ability to identify preference patterns that are not necessarily obvious from product attributes alone. Users may discover products through the behavior of other users with similar interests, even when the system cannot derive the relationship directly from technical specifications.

Nevertheless, collaborative filtering depends on the availability and quality of interaction data. New users and newly introduced products may have insufficient history for reliable recommendations. Sparse interaction matrices, changing preferences, and popularity bias can also affect recommendation quality.

These limitations are particularly relevant to a graduation project whose available catalogue and interaction data may be limited. A system should not assume that a sophisticated collaborative model will necessarily outperform simpler methods under limited-data conditions.

4.4 Knowledge-Based and Hybrid Recommendation

Knowledge-based recommendation is particularly relevant when users provide explicit requirements that must be satisfied. Such requirements may include a maximum budget, intended use, required technical features, or compatibility with existing equipment.

Rather than relying exclusively on historical behavior, a knowledge-based system can use explicit constraints and domain rules to evaluate candidate products.

Hybrid approaches combine techniques to benefit from their complementary strengths. Recent research continues to examine combinations of semantic matching, structured product relationships, and machine-learning-based methods in e-commerce recommendation. For example, Pua et al. (2026) investigated an ontology-based hybrid recommender that combines semantic matching with graph-based product relationships [8]. Its reported results are specific to the study's data and experimental setting and should not be generalized to TechWise without independent testing.

For TechWise, a hybrid design is a reasonable candidate for investigation because the platform's intended capabilities involve both explicit requirements and semantic interpretation. However, the specific algorithms and models should be selected based on the available data, technical constraints, and evaluation results rather than assumed in advance.

4.5 Implications for TechWise

The literature suggests that product discovery should not depend on a single recommendation technique in every situation.

A suitable conceptual flow is:

8. Interpret the user's stated intent.
9. Extract structured requirements from the request.
10. Validate the extracted fields and clarify essential ambiguities.
11. Retrieve candidate products from authoritative product records.
12. Apply hard constraints and deterministic business rules.
13. Rank eligible candidates using an explicitly defined scoring method.
14. Explain the reasons for the resulting recommendations.
15. Present the evidence and limitations relevant to the user's decision.

The first and seventh steps may involve AI-assisted language interpretation and explanation. Candidate retrieval, constraint enforcement, and factual validation should remain grounded in system data and deterministic logic.

This separation is important because a fluent recommendation can still be wrong if it relies on inaccurate product information or ignores a hard constraint.

5. Explainable AI and Trustworthy Recommendation

5.1 Why Explanation Matters

A recommendation provides a proposed outcome, but users may also need to understand why that outcome is appropriate.

In product purchasing, an explanation may identify the requirements a product satisfies, the compromises it introduces, or the reasons it ranks above another alternative. Such information can help users determine whether a recommendation reflects their actual priorities.

Explainability is not equivalent to generating a convincing paragraph. A useful explanation should be consistent with the evidence and decision process that produced the recommendation.

For example, if a product is recommended because it fits a budget and satisfies a required technical specification, the explanation should reflect those facts. It should not invent benchmark results, compatibility claims, or product characteristics that are absent from the underlying records.

5.2 Large Language Models for Recommendation Explanations

Said (2025) reviewed the use of large language models to explain recommendations. The review searched literature in the ACM Guide to Computing Literature for publications from the launch of ChatGPT through November 2024. Of 232 retrieved articles, six met the review's inclusion criteria as directly addressing LLM-based recommendation explanations [9].

This result indicates that, within the scope of that review, direct research on LLM-generated recommendation explanations was still limited. It should not be interpreted as a count of all explainable-AI research, nor as a current estimate of every relevant publication after the review's search period.

LLMs offer potential advantages in translating structured recommendation results into natural-language explanations. They can help users understand technical attributes, summarize trade-offs, and communicate the relationship between their requirements and candidate products.

However, language generation introduces additional risks. A generated explanation may contain unsupported details, overstate confidence, omit important trade-offs, or describe a recommendation as certain when the evidence is incomplete.

Accordingly, explanation quality should be evaluated independently of language fluency.

5.3 Separation of Interpretation and Decision Authority

A central architectural consideration for TechWise is the separation between AI interpretation and authoritative system decisions.

The AI layer may:

- Interpret natural-language shopping requests.
- Extract candidate requirements.
- Ask clarifying questions.
- Summarize verified product attributes.
- Explain results generated by the recommendation engine.
- Describe trade-offs between eligible alternatives.

Backend services should remain responsible for:

- Validating structured inputs.
- Retrieving authoritative product records.
- Enforcing hard constraints.
- Applying compatibility rules.
- Calculating decision scores.
- Checking resale eligibility and ownership authorization.
- Persisting transactions and lifecycle records.

This architecture reduces the risk of treating generated language as a substitute for data validation.

It also creates a clearer path for testing. If a recommendation is incorrect, the team can investigate whether the problem originated in intent extraction, product data, filtering, ranking, or explanation generation rather than treating the entire process as one opaque AI operation.

5.4 Evaluation of Explainability

A future TechWise evaluation should assess more than whether users find explanations readable.

Potential evaluation dimensions include:

- Factual consistency: Whether explanations accurately reflect product records and computed results.

- Constraint coverage: Whether important user requirements are addressed.
- Traceability: Whether a claim can be linked to an identifiable source or system calculation.
- Uncertainty communication: Whether missing or ambiguous information is disclosed.
- Usefulness: Whether explanations help users understand the recommendation.
- Decision support: Whether users can compare alternatives and identify trade-offs more effectively.

These dimensions are proposed evaluation criteria. Their measurement instruments, test cases, and acceptance thresholds must be defined and validated before results can be reported.

5.5 Implications for TechWise

AI-assisted shopping should be designed as a controlled interpretation and explanation layer over reliable product data and deterministic decision processes.

The literature supports investigating explainable recommendations, but it does not justify assuming that LLM-generated explanations are automatically faithful or that they necessarily improve purchasing outcomes.

TechWise should therefore prioritize grounded explanations that reference verified product attributes, explicit constraints, and recorded calculation results.

6. Product Information Quality and Product Truth

6.1 Product Information as a Decision Resource

Product information is one of the principal inputs to online purchasing decisions. Consumers use attributes, descriptions, images, and comparisons to assess whether an item satisfies their needs.

The quality of that information affects the usefulness of the decision process. A recommendation engine cannot reliably evaluate a specification that is absent, incorrectly recorded, or represented inconsistently across products.

Product information quality should therefore be treated as a system-wide concern rather than an isolated presentation issue.

6.2 Empirical Evidence on Product Information Quality

Research published in the International Journal of Retail & Distribution Management investigated the effects of product information quality using a scenario-based experiment involving 3,544 DIY online shoppers in the United Kingdom [3]. The study examined multiple information-quality criteria, levels of information quality, and product categories.

The findings indicated that poor product information quality negatively affected online shopping outcomes. The effects varied according to the information criterion, product category, and level of consumer involvement. The study also identified the importance of readable titles and comparable

product attributes, particularly for products involving higher consumer engagement.

Although the study offers empirical evidence concerning online product information, its results should be interpreted within its experimental context. DIY product categories and a United Kingdom sample do not establish identical effects for every electronics category or consumer population.

Nevertheless, the study supports a general design implication: the structure, completeness, and comparability of product information should be considered explicitly when developing systems that assist product selection.

6.3 Product Truth as a Platform Design Principle

For TechWise, the Product Truth Layer is a proposed architectural and business concept intended to ensure that product-related decisions are grounded in authoritative records rather than generated assumptions.

The concept requires a distinction between at least three forms of information:

Authoritative structured facts: Product attributes stored in platform records or supported by an identified authoritative source.

Seller-reported or user-provided information: Claims submitted by a seller or user, which may require validation or supporting evidence.

Derived information: Results produced through calculations, compatibility rules, or other deterministic processing of available data.

This classification is a proposed design model for TechWise. It does not imply that all data provenance mechanisms or verification workflows have already been implemented.

A practical implementation should preserve the origin of important claims and distinguish a missing value from a confirmed negative value. For example, an unknown compatibility status should not automatically be displayed as either compatible or incompatible.

6.4 Data Quality Dimensions

The following dimensions provide a useful starting point for evaluating product data.

Dimension	Meaning	Example
Accuracy	Whether a recorded value correctly represents the product	Correct processor model
Completeness	Whether required attributes are present	Display size is provided
Consistency	Whether attributes use compatible definitions and units	Storage capacities use a normalized unit
Comparability	Whether equivalent attributes can be compared across products	RAM capacity represented consistently
Timeliness	Whether the information remains current where change matters	Current availability or price

Dimension	Meaning	Example
Provenance	Whether the source of a claim is identifiable	Manufacturer specification versus seller claim
Uncertainty	Whether unknown or insufficiently supported values are distinguished	Compatibility not yet established

Table 2. Data quality dimensions relevant to the TechWise Product Truth Layer.

These dimensions are proposed quality criteria informed by the product information problem. They should be translated into specific validation rules for each supported product category.

6.5 Structured Product Data and Normalization

Different manufacturers and sellers may describe equivalent attributes using different names, units, or formats. A product intelligence system must normalize relevant fields before comparing them.

For example, two products may list memory capacity using different textual representations. A normalized representation enables consistent filtering and comparison, provided that the original meaning is preserved.

Normalization alone does not establish factual accuracy. A syntactically valid attribute may still be incorrect or unsupported. The platform therefore needs separate checks for data format, source reliability, and the business significance of each attribute.

Similarly, product-level specifications should not be confused with the condition or history of a specific physical device. A model's original specifications do not establish the current condition of an individual second-hand unit.

6.6 Implications for TechWise

Product Truth should be a foundational dependency for the platform's other core capabilities.

Recommendation, comparison, compatibility, decision scoring, and resale eligibility all depend on reliable information. Errors in the underlying product records may propagate into multiple features, even if each individual algorithm is functioning as intended.

Consequently, TechWise should establish category-specific data requirements, validation rules, provenance handling, and explicit representations of unknown information before claiming that the platform provides reliable product intelligence.

7. Product Compatibility and Semantic Product Relationships

7.1 Beyond Similarity-Based Product Matching

Two products can be similar without being compatible. Likewise, two products may have different categories but still need to be evaluated together because one depends on the other.

This distinction is important for electronic products, where suitability may depend on connectors, supported standards, operating requirements, dimensions, capacity, or other technical constraints.

A product discovery system based only on textual similarity may identify products that appear relevant but cannot satisfy a specific technical requirement. Compatibility evaluation therefore requires structured attributes and explicit relationships that represent the conditions under which products can work together.

7.2 Ontology-Based Product Recommendation

An ontology provides a formal representation of concepts, properties, and relationships within a domain. In product intelligence, an ontology can represent product categories, technical attributes, and relationships between products or components.

Alsobhi and Amare (2022) presented an ontology-based relational product recommendation system and discussed how product relationships can be used to support recommendations [10]. Their work provides a relevant foundation for considering relational knowledge in product recommendation, although the specific model and evaluation should not be assumed to generalize to every electronics compatibility problem.

Tiryaki and Yücebaş (2023) investigated an ontology-based personalized product recommendation system for e-retail, combining machine learning and semantic matching. Their study focused on retail categories including clothing, shoes, and bags, demonstrating a domain-specific application of semantic product relationships rather than establishing an electronics compatibility solution [11].

These studies support the relevance of structured semantic relationships, but they also illustrate why domain transfer requires care. Semantic similarity in clothing recommendations is not equivalent to electrical, mechanical, or software compatibility.

7.3 Compatibility as a Rule-Based Evaluation

A compatibility system should represent the attributes and conditions required to determine whether two products can work together.

A conceptual evaluation process may include:

16. Identifying the relevant products and their authoritative specifications.
17. Identifying the compatibility attributes required for the relationship.
18. Normalizing attribute values into comparable representations.
19. Applying explicit compatibility rules.
20. Returning a result supported by the available evidence.
21. Explaining the decisive attributes and any missing information.

The system should distinguish at least three outcomes:

- Compatible: Available data satisfies the defined compatibility rules.

- Incompatible: Available data violates at least one decisive rule.
- Unknown: Available data is insufficient to establish either outcome.

The three-state model is a proposed design recommendation for TechWise, not a claim that all existing compatibility systems use the same classification.

It is particularly important because missing information should not be interpreted as proof of compatibility.

7.4 Product Compatibility Graph

The Product Compatibility Graph is one of TechWise's confirmed core features. At the conceptual level, a graph can represent products or product entities as nodes and their relationships as edges.

However, not every product relationship should be treated as a compatibility relationship. Similarity, alternative selection, accessory association, dependency, and verified compatibility represent different meanings and should be modeled separately.

A robust design should define:

- The entity represented by each node.
- The meaning and direction of each relationship.
- The attributes required to establish the relationship.
- The evidence or rule supporting the relationship.
- The handling of incomplete or conflicting information.
- The conditions under which a relationship becomes outdated.

A graph representation can improve the organization and retrieval of product relationships, but a graph database alone does not guarantee accurate compatibility. The quality of the rules and underlying information remains decisive.

7.5 Implications for TechWise

The literature supports investigating semantic product representations and relational recommendation methods as foundations for product intelligence.

For TechWise, compatibility evaluation should be implemented as an explicit backend capability supported by structured specifications and defined rules. AI may help interpret a user's description of the equipment they own or explain the result, but it should not independently invent or certify a technical relationship.

The proposed approach should be validated against representative product categories, including cases with missing, contradictory, or borderline specifications.

8. Purchase Decision Support and Multi-Criteria Evaluation

8.1 The Purchase Decision as a Multi-Criteria Problem

Choosing a product often involves several competing criteria rather than a single objective. A consumer may consider price, performance, compatibility, durability, warranty, intended use, and future upgrade requirements.

An option with the highest performance may exceed the budget. A cheaper alternative may fail to satisfy a required specification. A third option may offer sufficient performance at a lower total cost.

A decision-support system must therefore distinguish mandatory constraints from preferences that can be traded off against each other.

This distinction is particularly important when the user describes requirements in natural language. The system must determine which conditions are essential and which represent preferences, asking for clarification when the distinction materially affects the result.

8.2 Hard Constraints and Soft Preferences

A useful conceptual model separates requirements into two categories.

Hard constraints define conditions an alternative must satisfy to remain eligible. Examples include a strict maximum budget or a required technical interface.

Soft preferences describe qualities that influence ranking but may be traded off against other considerations. Examples include a preferred brand, design preference, or desire for additional performance.

A candidate that violates a hard constraint should not automatically remain eligible merely because it scores highly on other criteria.

This approach provides a clear separation between filtering and ranking:

- Filtering determines which candidates satisfy the required conditions.
- Ranking orders the eligible candidates according to defined preferences and weights.
- Explanation communicates why a candidate appears in a particular position.

This separation is a proposed design approach for TechWise. Its suitability should be tested against representative user scenarios and edge cases.

8.3 Multi-Criteria Decision Analysis

Multi-criteria decision analysis (MCDA) provides a family of methods for evaluating alternatives across multiple criteria. Such methods can support explicit comparison of trade-offs, provided that the criteria, scales, weights, and aggregation rules are appropriately defined.

For example, a product decision engine could evaluate eligible products against criteria such as cost, performance, compatibility, and suitability for the user's intended workload.

However, assigning numerical scores requires care. Different criteria may use different units and scales, and the selection of weights can substantially influence the ranking. A weighted score should not be presented as an objective truth when it depends on subjective priorities or uncertain data.

The system should disclose relevant assumptions and avoid false precision. If two products are effectively indistinguishable given the available evidence, the output should not imply a meaningful difference solely because their calculated scores differ slightly.

8.4 Purchase, Upgrade, or Resell

TechWise extends decision support beyond selecting a new product. Its intended lifecycle also includes evaluating whether a user should retain an existing product, upgrade it, or consider reselling it.

This introduces a different decision problem from conventional product ranking.

The comparison may require information about the existing device, the user's current requirements, possible upgrade options, replacement cost, resale value, compatibility constraints, and the expected usefulness of each alternative.

Research on repair versus replacement provides relevant evidence concerning lifecycle decisions. Bovea, Ibáñez-Forés, and Pérez-Belis (2020) examined repair-and-reuse options against replacement for nine categories of small household electrical and electronic equipment. Their findings indicated that repair and reuse were generally environmentally preferable in the examined cases, although some repair scenarios favored replacement [12].

The study is useful because it demonstrates that lifecycle decisions can depend on product category and scenario. It does not establish that repair is always financially preferable, nor does it directly validate a consumer-facing buy-versus-upgrade algorithm.

TechWise should therefore avoid assuming that environmental benefit, financial value, and technical suitability always produce the same recommendation.

8.5 Decision Transparency and Sensitivity

A decision-support system should help users understand how a result changes when their priorities change.

For example, a consumer may prefer lower cost over additional performance, while another may prioritize future workload requirements. The ranking should reflect the specified priorities rather than a hidden universal definition of the best product.

A future evaluation of TechWise should investigate:

- Whether the selected criteria reflect user needs.

- Whether scores are reproducible from the recorded inputs.
- Whether hard constraints are consistently enforced.
- Whether changing a preference produces an understandable change in ranking.
- Whether missing data affects the result appropriately.
- Whether explanations accurately describe trade-offs.

These are proposed evaluation objectives, not completed test results.

8.6 Implications for TechWise

The Purchase Decision Engine should provide structured and explainable comparisons rather than relying on an unconstrained AI judgment.

The AI layer can assist with requirement extraction and communication, while deterministic services apply constraints, calculate scores, and produce traceable decision results.

The Buy vs Upgrade vs Resell Engine should use lifecycle-aware inputs and clearly distinguish financial, technical, and environmental considerations. Where the available data is insufficient, the system should communicate uncertainty instead of presenting a definitive recommendation.

9. Second-Life Commerce, Trust, and Product Lifecycle Traceability

9.1 Trust in Second-Hand Electronic Commerce

Second-hand commerce introduces uncertainties that may be less prominent in the purchase of a new product from an established retailer. Consumers may have limited knowledge of the item's previous use, current condition, maintenance history, seller reliability, or authenticity.

These uncertainties affect the information available to the buyer and the confidence with which the product can be evaluated.

Lee and Lee (2005) examined initial trust in the online second-hand product market using an experimental setting involving used notebook computers. Their study involved 163 college students and investigated the relationship between trust in the store, trust in the product, and willingness to purchase [13]. The results indicated that store-level and product-level trust were distinct considerations and that both were relevant to purchase intentions in the studied setting.

The study offers a useful conceptual distinction between confidence in the platform or seller and confidence in the individual product. However, its age, sample, and experimental design limit the extent to which its findings can be generalized to contemporary resale platforms or all consumer groups.

The distinction remains relevant to TechWise: a trustworthy marketplace does not automatically establish the physical condition of every listed item, and evidence about an individual item does not automatically establish the reliability of the entire marketplace.

9.2 Evidence, Claims, and Verification

A second-life marketplace needs to distinguish what a seller reports from what the platform can substantiate.

For example, a seller may report that an electronic device is fully functional. Supporting evidence may include photographs, a test video, or documented test results. The existence of that evidence does not necessarily mean that the platform has independently inspected or physically verified the device.

A proposed evidence model for TechWise distinguishes:

- Seller-reported information: Claims submitted by the current owner or seller.
- Supporting evidence: Materials or records that support a specific claim.
- Platform validation: Checks performed by the platform according to defined rules.
- Platform confidence: A status derived from the available evidence and validation process.

These categories must not be conflated. A platform should not label an item as physically verified unless an actual verification procedure has been implemented and completed.

This distinction is particularly important for electronics because a device may appear functional in a video while still having defects that are not observable through that evidence.

9.3 Product Lifecycle and Ownership

TechWise treats the product lifecycle as a continuous sequence of discovery, purchase, ownership, use, upgrade or resale, and possible transfer to a new owner.

A key modeling requirement is the distinction between a product model and a specific physical device.

Product refers to the model or SKU and its general specifications.

Product Asset refers to an individually identifiable physical device.

Ownership refers to a particular user's ownership period for that asset.

These concepts should not be collapsed into a single database record because they represent different kinds of information and change at different times.

For example, when a physical device is resold, its ownership changes, but its original manufacturing identity and product age do not reset. The new owner begins a new ownership period while the asset retains its previous lifecycle history.

A lifecycle-aware platform must also protect previous owners' personal information. Historical continuity does not justify exposing private account details or unrestricted transaction histories to subsequent owners.

These are established TechWise modeling decisions. Their implementation will require appropriate database relationships, authorization rules, lifecycle validation, and privacy controls.

9.4 Digital Product Passports and Lifecycle Information

Research on digital product passports (DPPs) investigates mechanisms for making product information more accessible across value chains and lifecycle stages.

Wagner (2025) examined information barriers to circularity for electronic products and discussed the potential and limitations of digital product passports in addressing those barriers [4]. The study is relevant because lifecycle decisions depend not only on the availability of product information, but also on whether the information can be accessed and used appropriately by different stakeholders.

Research on smart tags and digital product passports in circular electronics has also examined the potential role of item-level identification and lifecycle information in supporting decisions involving reuse, repair, remanufacturing, and recycling [5].

These studies suggest that lifecycle information can support decision-making when its structure, accessibility, and relationship to the physical product are clearly defined.

However, a digital record is not automatically proof that every recorded claim is accurate. Lifecycle information systems still depend on the quality of data entry, the reliability of contributing sources, and the mechanisms used to validate claims.

9.5 Traceability and Data Governance

For TechWise, lifecycle traceability should establish which asset is being discussed, which ownership period is active, and which records support a relevant claim.

A conceptual lifecycle sequence is:

Product → Product Asset → Purchase → Ownership Period → Usage and Evidence → Resale Eligibility → New Ownership Period

The sequence describes a proposed data and business model, not a claim that every stage has already been implemented.

The system should preserve the distinction between immutable historical events and mutable current state. For example, the active owner may change, while the original purchase event and earlier ownership records remain part of the lifecycle history.

Access to those records must be controlled. A new owner may need relevant information about the asset's age, purchase origin, and documented condition without receiving the previous owner's personal account information.

9.6 Second-Life Eligibility Rules

The confirmed TechWise concept restricts second-life eligibility to products satisfying the platform's purchase/origin and ownership conditions.

The proposed eligibility process should check:

- Whether the product asset has a valid platform purchase or recognized origin record.
- Whether the requesting seller is the current authorized owner.
- Whether the ownership period is active.
- Whether the asset is eligible for resale.
- Whether an existing listing or blocking dispute prevents the transaction.
- Whether the required evidence has been submitted.
- Whether the platform's eligibility rules are satisfied.

The exact implementation details, evidence requirements, and state transitions should be documented in the formal requirements and business-rule specifications.

A resale reference code can help connect the listing to the relevant asset and purchase history, but the code alone should not be treated as proof of current ownership. Authorization must be checked against the authoritative platform records.

9.7 Implications for TechWise

The literature supports the relevance of trust, information availability, and lifecycle traceability to second-hand commerce. It does not establish that any single evidence type can guarantee product condition or eliminate fraud.

TechWise should therefore combine lifecycle-aware data modeling, controlled access, deterministic eligibility checks, and clearly labeled evidence status.

Its second-life marketplace should distinguish the identity of the product asset from the identity of its current owner, preserve the asset's historical continuity, and communicate the limits of the platform's verification process.

10. Synthesis and Comparative Analysis

10.1 Cross-Domain Comparison

The reviewed research areas address complementary aspects of the product lifecycle. Their relationship to TechWise can be summarized as follows.

Research domain	Contribution to the problem	Important limitation	TechWise implication
Recommender systems	Reduce information overload and support product discovery	Recommendations depend on data, context, and method suitability	Combine explicit requirements with appropriate ranking methods
Explainable AI	Communicate reasons behind recommendations	Fluent explanations may not be faithful to underlying evidence	Generate explanations from validated facts and computed results
Product information quality	Improves the usefulness and comparability of product data	Information can be missing, inconsistent, or inaccurate	Establish structured attributes, validation, and provenance
Semantic product	Represent relationships	Similarity does not establish	Use explicit compatibility

Research domain	Contribution to the problem	Important limitation	TechWise implication
relationships	between products and concepts	technical compatibility	relations and deterministic rules
Decision-support methods	Organize trade-offs across multiple criteria	Weights and scoring assumptions affect results	Separate eligibility, ranking, and explanation
Second-hand commerce research	Explains the role of trust and uncertainty	Trust in the seller does not prove individual item condition	Distinguish seller claims, evidence, and platform validation
Lifecycle information systems	Support traceability and lifecycle decisions	Records may be incomplete or difficult to validate and share	Preserve asset history while controlling access and ownership transitions

Table 3. Cross-domain synthesis — contributions, limitations, and TechWise implications.

The comparison shows that no single research domain addresses all the requirements of TechWise. The proposed platform depends on combining the strengths of these approaches while managing their distinct limitations.

10.2 Shared Technical Dependencies

Several dependencies recur across the research domains.

First, recommendation quality depends on product information quality. A ranking method cannot compensate reliably for missing or inaccurate specifications.

Second, compatibility and decision support require structured data and explicit rules. Natural-language interpretation may help extract requirements, but authoritative validation must occur against the underlying records.

Third, explainability depends on traceability. A recommendation is easier to justify when the system can identify the constraints, product attributes, and calculations that influenced the result.

Fourth, lifecycle decisions require persistent identity and history. A second-hand product cannot be evaluated reliably using only the generic specifications of its model when its individual condition and ownership history also matter.

Finally, privacy and authorization are cross-cutting requirements. Lifecycle continuity should preserve relevant product history without granting subsequent owners unrestricted access to previous owners' personal information.

10.3 Architectural Implications

The findings support a layered conceptual architecture for TechWise.

User Experience Layer

Search · Product Discovery · Comparison · Decision Support · Ownership · Second-Life

AI Assistance Layer

Intent Interpretation · Requirement Extraction · Explanations · Summarization

Deterministic Business and Intelligence Services

Validation · Filtering · Ranking · Compatibility · Decision Calculations · Resale Eligibility · Authorization

Authoritative Data Layer

Product Records · Product Assets · Ownership History · Purchases · Evidence · Lifecycle Events

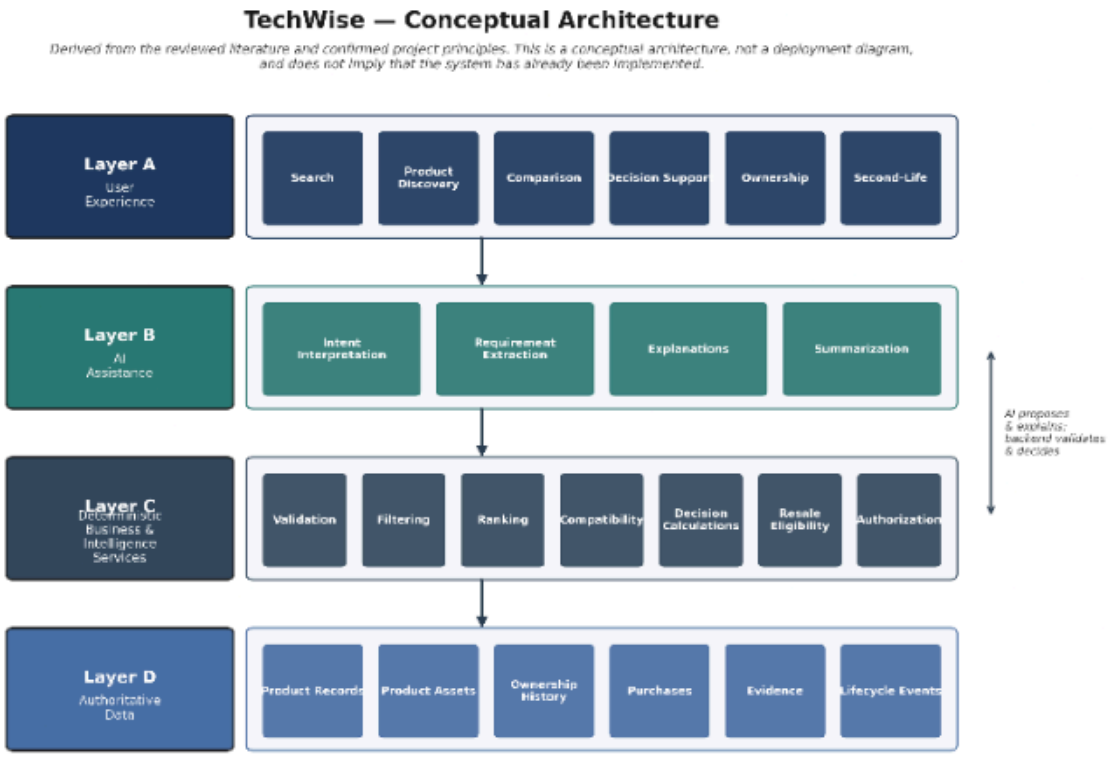


Figure 1. TechWise conceptual architecture — four layers separating user experience, AI assistance, deterministic services, and authoritative data.

Figure 1. TechWise conceptual architecture — four layers separating user experience, AI assistance, deterministic services, and authoritative data. Derived from the reviewed literature and confirmed project principles; not a deployment diagram or a claim of completed implementation.

Conceptual architecture derived from the literature and confirmed project principles; not a deployment diagram or a claim of completed implementation.

This model places AI in a supporting role and keeps factual decisions tied to validated data and backend logic. It also provides clear boundaries for testing individual components and tracing errors through the decision process.

11. Research Gaps and Motivation for TechWise

11.1 Fragmentation Across Product Decision Stages

The reviewed studies address different parts of the consumer's product journey. Recommender systems focus on discovering relevant items, explainable recommendation investigates the communication of recommendation logic, and lifecycle research considers information needed across stages such as reuse and resale.

These approaches are complementary, but their existence does not prove that they are commonly integrated into one consumer-facing platform with consistent rules and persistent lifecycle records.

Implication for TechWise: The project proposes to connect these concerns through a shared product intelligence and lifecycle model.

This is an integration objective, not a claim that no existing platform or research prototype offers comparable capabilities. Establishing the uniqueness of the integration would require a broader comparative review of commercial systems and research implementations.

11.2 Gap Between Natural-Language Intent and Validated Product Decisions

Natural-language interfaces can make product discovery more accessible, but users' descriptions must ultimately be translated into structured requirements that a system can validate.

An AI-generated interpretation may be incomplete or ambiguous. If the platform proceeds without identifying essential missing information, it may return irrelevant recommendations or unsupported compatibility claims.

Implication for TechWise: The proposed workflow separates intent interpretation, structured extraction, validation, deterministic processing, and result explanation.

Future evaluation should measure extraction accuracy, constraint adherence, clarification behavior, and the frequency of unsupported claims.

11.3 Insufficient Reliability When Product Data Is Incomplete

Recommendation, compatibility, and decision-support mechanisms all depend on the quality of their inputs. A product record may be incomplete even when its available fields are internally consistent.

Implication for TechWise: The Product Truth Layer should distinguish verified facts, seller-reported claims, derived values, and unknown information.

A research and testing priority is to determine whether these distinctions reduce unsupported product claims and improve the reliability of downstream decisions.

11.4 Compatibility Requires More Than Product Similarity

Semantic similarity can identify related products, but technical compatibility depends on specific attributes and constraints.

Implication for TechWise: The Product Compatibility Graph should represent explicit relationship types and use backend validation rules to determine compatibility.

The system should be evaluated against known-compatible, known-incompatible, and insufficient-information cases rather than relying solely on recommendation relevance metrics.

11.5 Decision Support Requires Explicit Trade-Offs

A product recommendation may not explain why an alternative is preferable under a user's specific budget, performance needs, or existing equipment constraints.

Similarly, buy-versus-upgrade-versus-resell decisions require information about the current asset and the alternatives, not simply a generic product ranking.

Implication for TechWise: The Purchase Decision Engine and Buy vs Upgrade vs Resell Engine should expose their relevant criteria, enforce mandatory constraints, and provide explanations that reflect the actual decision process.

Their usefulness must be tested with realistic scenarios and user feedback.

11.6 Second-Life Trust Requires Lifecycle-Aware Evidence

Research on second-hand commerce identifies trust as an important consideration, while lifecycle information research highlights the importance of traceability. Neither establishes that a seller's claim, an uploaded video, or a digital record alone proves the physical condition of an individual device.

Implication for TechWise: The Verified Second-Life Marketplace should apply explicit eligibility rules, preserve asset history, distinguish claims from evidence, and accurately represent the scope of platform validation.

A future evaluation should examine the consistency of eligibility decisions, the handling of invalid ownership attempts, and the clarity of evidence status presented to buyers.

11.7 Privacy Must Coexist With Lifecycle Continuity

Persistent product history is valuable for lifecycle decisions, but unrestricted historical access may expose personal information belonging to previous owners.

Implication for TechWise: Ownership history and asset history should be represented separately, with authorization rules that expose only the information necessary for each operation.

The system should be evaluated for unauthorized access and privacy leakage as well as functional correctness.

11.8 Consolidated Research Gap

The reviewed literature establishes relevant foundations for recommendation, explainability, information quality, semantic product relationships, decision support, second-hand trust, and lifecycle traceability. The central motivation for TechWise is to investigate how these capabilities can be coordinated within one product lifecycle platform while maintaining reliable data, explicit decision rules, traceable explanations, and controlled ownership transitions.

The gap should be interpreted as a design and integration problem identified from the reviewed sources, not as proof that no comparable integrated system exists. A stronger novelty claim would require systematic comparative analysis of existing platforms, research prototypes, and their actual capabilities.

12. Proposed Conceptual Approach for TechWise

12.1 Platform Overview

TechWise is proposed as a Smart E-Commerce and Product Intelligence Platform that supports the consumer beyond conventional product browsing and purchasing.

The platform's lifecycle vision connects product discovery and evaluation with purchasing, ownership, upgrading, resale, and ownership transfer. This lifecycle perspective provides a common context for the platform's six confirmed core features.

The proposed approach uses AI for natural-language interpretation and explanation while retaining deterministic backend services as the authority for validation, business rules, product facts, compatibility, authorization, and lifecycle state.

12.2 AI Shopping Intelligence

This capability focuses on interpreting user intent and assisting consumers in finding relevant products.

The AI layer may extract requirements from natural-language requests, identify missing information, and explain candidate products. The backend then validates the extracted requirements, retrieves candidate products, applies constraints, and returns the resulting recommendations.

The design should distinguish explicit requirements from inferred preferences and should not treat an uncertain interpretation as a confirmed user decision.

12.3 Product Compatibility Graph

This capability represents meaningful relationships between products and supports compatibility evaluation.

Its effectiveness depends on structured specifications, defined relationship semantics, and explicit

validation rules. AI can help users describe their existing equipment and understand compatibility results, but compatibility outcomes must remain grounded in product records and the platform's defined rules.

12.4 Purchase Decision Engine

This capability supports comparison and evaluation across multiple criteria.

The proposed design separates hard-constraint filtering from ranking and explanation. It should communicate material trade-offs and distinguish measured product attributes from subjective user preferences.

Any scoring method should be documented and evaluated before its outputs are treated as reliable decision guidance.

12.5 Product Truth Layer

This capability provides a conceptual foundation for trustworthy product information.

It should support structured product attributes, consistent representations, identifiable information sources, and clear distinctions between confirmed values, claims, derived values, and unknowns.

Its role is cross-cutting because the reliability of recommendation, compatibility, comparison, and resale eligibility depends on the underlying information.

12.6 Verified Second-Life Marketplace

This capability supports eligible resale transactions for products that satisfy the platform's purchase/origin and ownership conditions.

The proposed process validates the current ownership state and other eligibility rules before allowing a listing to proceed. Required evidence should be handled according to explicit rules, without treating submitted evidence as proof of physical verification unless the platform actually performs that verification.

Lifecycle history should remain associated with the physical asset while ownership periods change, and previous owners' personal information must remain protected.

12.7 Buy vs Upgrade vs Resell Engine

This capability extends decision support to the consumer's existing product and possible next actions.

Its proposed evaluation may consider the user's current requirements, available upgrade options, replacement alternatives, relevant costs, compatibility constraints, and resale information.

The system should distinguish financial value, technical suitability, and environmental considerations rather than combining them into an unexplained universal score. Recommendations must reflect the data available and disclose material uncertainties.

12.8 Research and Evaluation Priorities

The conceptual approach generates several testable research and engineering questions.

Evaluation area	Example question	Potential evidence
Requirement extraction	Does the system accurately extract explicit user constraints?	Labeled test requests and extraction results
Recommendation	Do returned products satisfy mandatory constraints?	Constraint-validation test cases
Explanation	Are generated explanations supported by actual product facts?	Fact-to-explanation consistency review
Information quality	Are required product attributes complete and comparable?	Data-quality checks
Compatibility	Are known-compatible and incompatible cases classified correctly?	Reference compatibility test set
Decision support	Are rankings reproducible and consistent with specified priorities?	Controlled scoring tests
Second-life eligibility	Are invalid ownership and ineligible listings rejected?	Authorization and business-rule tests
Lifecycle integrity	Are asset history and ownership transitions preserved correctly?	Lifecycle integration tests
Privacy	Can users access only authorized ownership information?	Access-control and privacy tests
User usefulness	Do users understand recommendations and trade-offs?	Usability evaluation and participant feedback

Table 4. Proposed TechWise research and evaluation priorities.

These are proposed evaluation methods. They are not reported experimental results, and no performance figures should be included until the team has conducted the corresponding tests.

13. Limitations of the Review

Several limitations should be considered when interpreting the findings.

First, the current review uses a focused selection of relevant publications rather than an exhaustive systematic search. Additional research may identify other approaches, contradictory findings, or existing platforms with more integrated capabilities than those discussed here.

Second, the reviewed studies use different methods, datasets, product categories, and evaluation metrics. Their findings cannot be compared as if they were generated under a common experimental design.

Third, some sources concern general e-commerce recommendation or non-electronics retail categories. Their methods may be informative, but their effectiveness for technical product compatibility must be established independently.

Fourth, research on second-hand trust and lifecycle traceability does not prove that the proposed TechWise evidence model will prevent fraud or establish the physical condition of every listed

product.

Fifth, the architectural approach presented in this document is a proposal derived from the literature and the project's confirmed design principles. Its effectiveness remains dependent on implementation quality, available data, security controls, and empirical evaluation.

Finally, the novelty of TechWise's integrated approach has not been conclusively established. A comprehensive novelty assessment would require a more extensive review of existing research systems and commercial platforms.

These limitations define the appropriate interpretation of this document: it is an evidence-informed foundation for system design and subsequent validation, not proof of completed implementation or demonstrated superiority.

14. Conclusion

This literature review examined the research domains relevant to TechWise as a Smart E-Commerce and Product Intelligence Platform. The analysis covered recommender systems, explainable AI, product information quality, semantic product relationships, multi-criteria decision support, second-hand commerce, and product lifecycle traceability.

The reviewed literature demonstrates that these domains address different but interdependent aspects of product selection and ownership. Recommender systems can support discovery, but their usefulness depends on the quality of the underlying data and the appropriateness of their methods. AI-generated explanations can improve communication, but they must remain faithful to the evidence supporting a recommendation. Product compatibility requires explicit technical rules rather than similarity alone. Decision support requires clear constraints and documented trade-offs. Second-life commerce requires trust-aware handling of evidence, ownership, and product history.

These findings motivate TechWise's proposed integrated approach. Its six core features are intended to connect intelligent shopping assistance, product truth, compatibility evaluation, purchase decision support, second-life commerce, and lifecycle-aware upgrade or resale decisions.

The principal architectural implication is that intelligence and authority should remain distinct. AI can help users express their needs and understand the results, while deterministic backend services and authoritative records govern factual validation, business rules, compatibility, authorization, and lifecycle state.

The review therefore provides a foundation for the next stages of requirements specification, system analysis and design, implementation, and testing. Future work should expand the literature search, validate the proposed architecture, define measurable acceptance criteria, and evaluate the resulting system using reproducible tests and user-centered evidence.

TechWise's contribution should ultimately be assessed not by the number of technologies it

combines, but by whether the integration enables consumers to make better-informed decisions while maintaining reliable product information, transparent reasoning, secure ownership transitions, and appropriately qualified claims about second-life products.

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